

1 Introduction

Frequency-selective and impedance-matching networks are fundamental building blocks in RF and microwave systems. Unlike the frequency-independent transmission lines studied in Lab 2, this assignment focuses on components whose performance is inherently frequency-dependent. Two categories of design are explored:

- (1) lumped LC matching networks that transform the complex input impedance of a CPW line to $50\ \Omega$ at 550 MHz and 5 GHz.
- (2) distributed microstrip multi-port devices comprising a Wilkinson power divider at 1 GHz and a two-way diplexer separating 1 GHz and 2 GHz signals.

All designs use the same PCB substrate ($\epsilon_r = 2.94$, $h = 0.076$ mm, copper conductor) and are simulated in Keysight ADS using closed-form microstrip and CPW models.

2 EDA Tool Use

All simulations use the S-parameter controller in ADS with frequency sweep parameters chosen to resolve the key features of each design. Table 1 summarises the sweep configurations. The substrate is defined via the MSUB block with $H = 0.076$ mm, $\epsilon_r = 2.94$, $T = 0.035$ mm, $\tan \delta = 0$. MTEE junction components are used at all T-junctions to account for current redistribution and fringing capacitance discontinuities. Lumped L and C components are used for the matching networks; MLIN and MLOC components are used for all distributed designs.

Table 1: S-parameter simulation sweep setup for each device in ADS.

Device	Start (GHz)	Stop (GHz)	Step (MHz)
CPW Matching Network (550 MHz)	0.1	1.0	1
CPW Matching Network (5 GHz)	4.0	6.0	5
Wilkinson Power Divider	0.1	2.0	1
Diplexer	0.1	3.0	1

Table 2: Theoretical and ADS-tuned lumped element values for CPW matching networks. The Wilkinson divider and diplexer use only distributed microstrip lines with no lumped components.

Device	f_0	Theoretical		Practical (Tuned)		Topology
		L (nH)	C (pF)	L (nH)	C (pF)	
CPW Matching Network	550 MHz	19.30	5.79	19.30	2.895	Series L , Shunt C
CPW Matching Network	5 GHz	2.48	0.202	2.48	0.2016	Shunt C , Series L

Derivation of Theoretical Lumped Element Values

Lumped element values are derived using the L-network method. For source $Z_S = 50\ \Omega$ and load $Z_L = R_L + jX_L$, the network quality factor is:

$$Q = \sqrt{\frac{R_{\text{high}}}{R_{\text{low}}} - 1} \quad (1)$$

At 550 MHz: $Z_L = 25 - 41.7j \Omega$ (extracted from ADS Smith chart). Since $R_L < Z_S$, shunt element goes at source side:

$$Q = \sqrt{\frac{50}{25} - 1} = 1.0, \quad L_{\text{series}} = \frac{QZ_S + |X_L|}{2\pi f_0} = \frac{91.7}{2\pi \times 550 \times 10^6} = 19.30 \text{ nH}, \quad C_{\text{shunt}} = \frac{1}{2\pi f_0 Q Z_S} = 5.79 \text{ pF} \quad (2)$$

The tuned $C = 2.895 \text{ pF}$ is half the theoretical value, attributed to $\approx 2.895 \text{ pF}$ parasitic capacitance inherent to the CPW geometry.

At 5 GHz: $Z_L = 140.1 - 66.2j \Omega$. Since $R_L > Z_S$, shunt element goes at load side. Solving the admittance matching condition $B_C^2 + 2B_L B_C + B_L^2 + G^2 - G/Z_0 = 0$ gives:

$$C_{\text{shunt}} = 0.202 \text{ pF}, \quad L_{\text{series}} = 2.48 \text{ nH} \quad (3)$$

Theoretical and tuned values agree at 5 GHz, confirming CPW parasitics are negligible at this frequency.

3 Results – Matching Networks

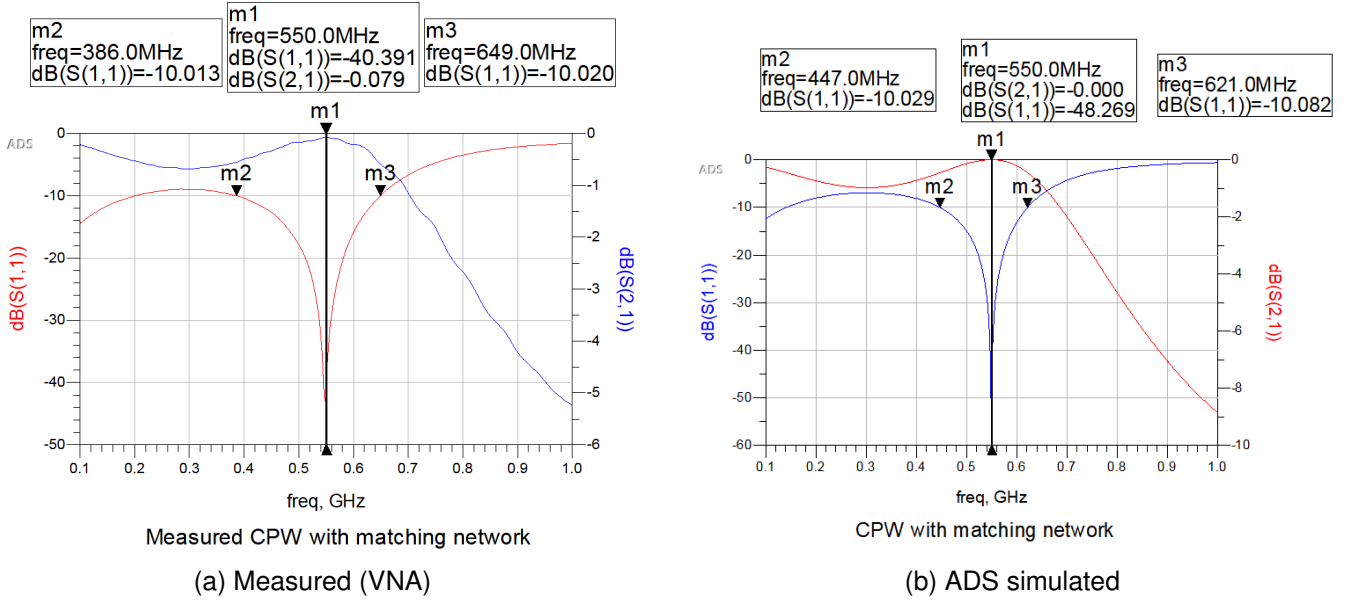


Figure 1: S-parameter response of the CPW line with LC matching network at 550 MHz. (a) VNA measurement: $|S_{11}| = -40.4 \text{ dB}$, 10 dB return loss bandwidth 386–649 MHz. (b) ADS simulation: $|S_{11}| = -48.3 \text{ dB}$, bandwidth 447–621 MHz. Both confirm deep matching at 550 MHz with $|S_{21}| \approx 0 \text{ dB}$ confirming lossless network. Discrepancy between measured and simulated bandwidths arises from connector parasitics and fabrication tolerances not captured in the ideal ADS model.

The 550 MHz matching network achieves $|S_{11}| = -48.3 \text{ dB}$ in simulation and -40.4 dB in measurement, both representing near-perfect matches. The 4.8 dB discrepancy is consistent with connector and SMA launch parasitics which introduce additional reactive loading not modelled in ADS. The 10 dB return loss bandwidth (386–649 MHz simulated) corresponds to a fractional bandwidth of 47.8%, typical for an L-network with $Q = 1.0$.

At 5 GHz, the network achieves $|S_{11}| = -49.1 \text{ dB}$ with a narrower fractional bandwidth of 6.8%, consistent with the higher transformation ratio ($R_L/Z_S = 2.8$, $Q = 1.34$). The topology reversal – shunt element at the load side rather than source side – is required because the CPW real impedance (140.1Ω) exceeds $Z_S = 50 \Omega$ at 5 GHz, opposite to the 550 MHz case.

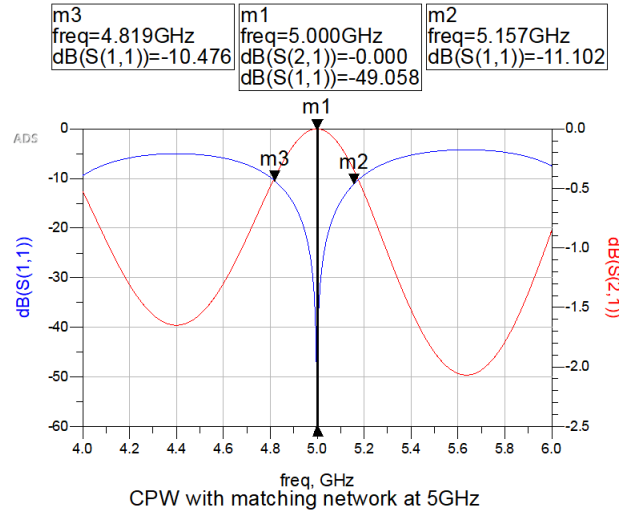


Figure 2: ADS simulated S-parameters of CPW with LC matching network at 5 GHz. $|S_{11}| = -49.1$ dB at 5 GHz and $|S_{21}| \approx 0$ dB confirm near-perfect impedance match. The 10 dB return loss bandwidth spans 4.82–5.16 GHz (340 MHz, 6.8% fractional bandwidth), narrower than the 550 MHz network due to the higher Q transformation ratio ($Q = 1.34$ vs $Q = 1.0$).

4 Results – Multi-Port Device Design

Wilkinson Power Divider

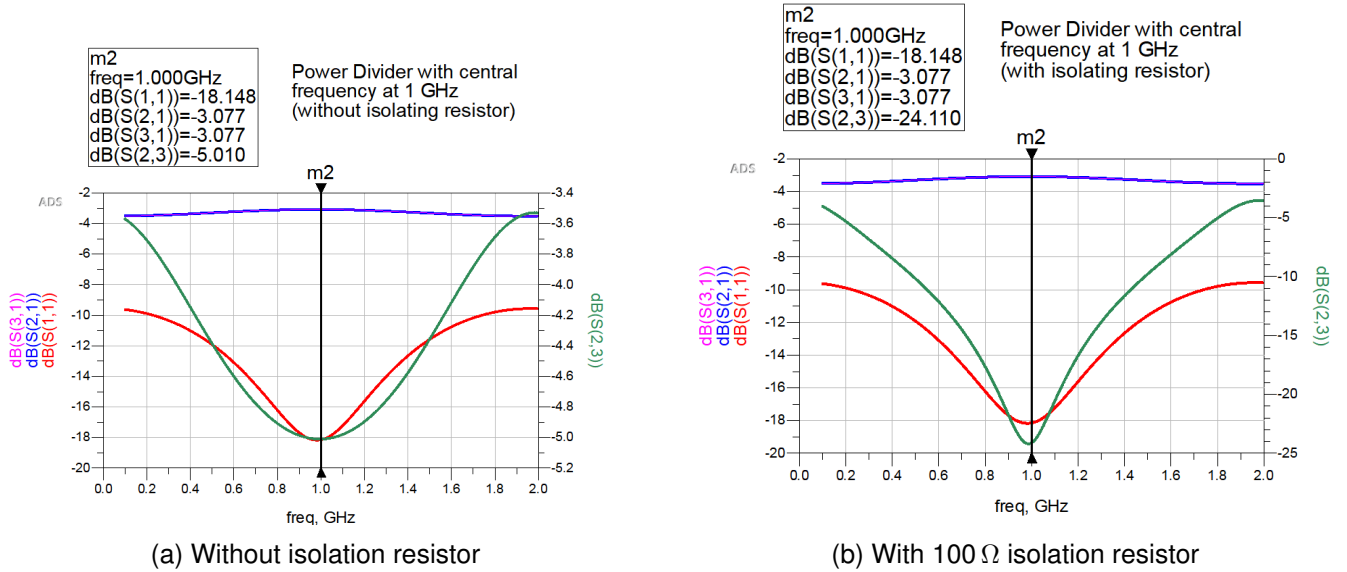


Figure 3: S-parameter response of the Wilkinson power divider at 1 GHz ($W_{\text{arms}} = 0.1086$ mm, $L_{\text{arms}} = 51.5$ mm, $W_{\text{feed}} = 0.1953$ mm). (a) Without resistor: $|S_{21}| = |S_{31}| = -3.077$ dB, $|S_{23}| = -5.010$ dB, $|S_{11}| = -18.15$ dB – isolation is poor and output matching degrades. (b) With 100 Ω resistor: $|S_{21}| = |S_{31}| = -3.077$ dB (unchanged), $|S_{23}| = -24.1$ dB (19 dB improvement), confirming the resistor's sole role is port isolation. The 10 dB return loss bandwidth spans 938–1062 MHz (124 MHz, 12.4% fractional bandwidth).

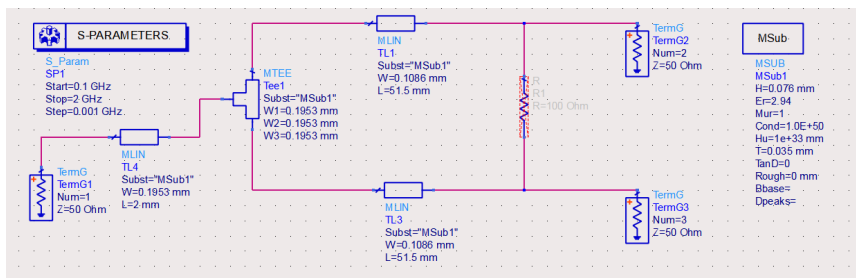
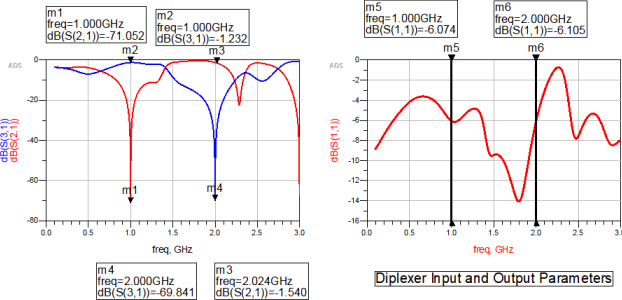
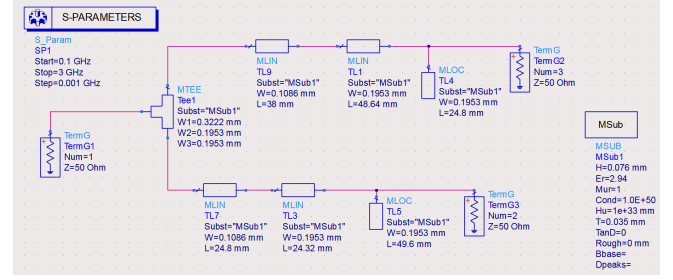


Figure 4: Wilkinson Power Divider setup without isolating resistor

Diplexer



(a) Transmission: $|S_{21}|$ and $|S_{31}|$



(b) ADS schematic

Figure 5: Two-way microstrip diplexer using open-ended $\lambda/4$ shunt stubs for band-stop filtering. Port 2 passes 1 GHz ($|S_{31}| = -1.23$ dB) and rejects 2 GHz ($|S_{21}| = -71.1$ dB > 20 dB spec). Port 3 passes 2 GHz ($|S_{21}| = -1.54$ dB) and rejects 1 GHz ($|S_{31}| = -69.8$ dB). Input VSWR < 3 at both bands ($|S_{11}| = -6.07$ dB at 1 GHz, -6.11 dB at 2 GHz). All four specifications are met. Isolation lines ($\lambda/4$ at each band frequency) prevent cross-path stub loading. Each arm uses a 70.7Ω $\lambda/4$ transformer for input matching.

5 Analysis and Discussion

Matching Network: Component Choice and Layout Considerations

The L-network topology using a series inductor and shunt capacitor was chosen for its simplicity – requiring only two components – and because it provides the minimum insertion loss for a given transformation ratio at a single frequency [1]. Alternative matching topologies include π and T-networks, which offer additional degrees of freedom to control bandwidth and Q independently, and single-section $\lambda/4$ transmission line transformers, which are lossless and wideband but only practical for purely resistive loads. At 550 MHz, lumped LC components are preferred over distributed stubs because the required $\lambda/4$ stub length (≈ 48 mm) is impractically large for a compact PCB layout. At 5 GHz, distributed stub matching becomes increasingly competitive since lumped component parasitics degrade performance, and $\lambda/4$ stubs shrink to ≈ 10 mm.

Practical layout considerations significantly affect matching network performance. Series inductors must be kept short with minimal via inductance; shunt capacitors require short, low-impedance paths to the ground plane to minimise parasitic series inductance that shifts the resonant frequency. Component placement close to the CPW port minimises the electrical length of interconnecting traces, which act as unintended series inductors. At 5 GHz, even a 1 mm trace adds ≈ 0.17 nH – comparable to the 2.48 nH design value – making layout parasitics non-negligible. Surface-mount 0402 or 0201 package components are recommended to minimise parasitic inductance in practical implementations [3].

Measured vs. Modelled S-Parameters

The measured $|S_{11}|$ at 550 MHz (-40.4 dB) differs from the ADS simulated value (-48.3 dB) by 7.9 dB. This discrepancy arises primarily from SMA connector parasitics (typically 0.1–0.3 nH series inductance and 0.1–0.5 pF shunt capacitance), PCB launch pad dimensions, and fabrication tolerances in trace width ($\pm 5\%$). For design purposes, measured S-parameters from a VNA provide a more reliable load characterisation than closed-form ADS models, as they inherently capture all parasitic effects including surface roughness, radiation, and substrate inhomogeneity. Designing a matching network directly from measured Z_{in} values – as done in this lab – is therefore more accurate than using modelled parameters, provided the measurement calibration is performed at the reference plane closest to the device under test.

Multi-Port Network: Practical Limitations and Error Sources

Three principal error sources were identified in the multi-port designs. **(1) T-junction discontinuities:** the Wilkinson divider required arm lengths of 51.5 mm rather than the analytically calculated 49.6 mm

(3.8% error), caused by fringing capacitance at the MTEE junction that adds electrical length. This effect scales with frequency and becomes critical at mmWave. **(2) Diplexer stub cross-coupling:** without $\lambda/4$ isolation lines, the band-stop stub on one path loaded both output ports simultaneously, as the short circuit presented by the stub at its resonant frequency propagated through the junction to the other path. **(3) Input impedance mismatch:** the parallel combination of two $50\ \Omega$ arms presents $25\ \Omega$ at the input junction, requiring $70.7\ \Omega\ \lambda/4$ arm transformers. Without these, VSWR exceeded 7, far beyond the specification of $\text{VSWR} < 3$.

Comparing the diplexer simulation to Line 3 (matched microstrip with open-ended $\lambda/4$ stub) on the test PCB confirms the band-stop mechanism – the stub creates a transmission null at the frequency where its length equals $\lambda/4$. The simulated null depth ($>60\ \text{dB}$) exceeds what is measured on Line 3 due to conductor and dielectric losses in the physical stub that broaden and shallow the null. In a real diplexer, this limits the achievable out-of-band rejection to typically 20–30 dB rather than the ideal $>60\ \text{dB}$ obtained in lossless ADS simulation.

Specification Limits and Microwave Frontend Impact

The Wilkinson divider 10 dB return loss bandwidth (938–1062 MHz, 12.4%) is fundamentally limited by the $\lambda/4$ arm length, which satisfies the impedance transformation condition only at exactly f_0 . For wider bandwidth, multi-section Wilkinson designs or coupled-line structures are used in practice [1]. The $\text{VSWR} < 3$ specification for the diplexer corresponds to a reflection coefficient of 0.5 ($|S_{11}| < -6.02\ \text{dB}$), which in a microwave frontend would cause up to 25% of transmit power to be reflected – acceptable for a receive path but potentially damaging for a high-power transmit PA. In LNA design, a poor input match degrades the noise figure by increasing the reflected signal and reducing power transfer from the antenna. For PA outputs, a $\text{VSWR} > 2$ can cause load-pull effects, compressing gain and reducing efficiency. This highlights why multi-port matching specifications are critical system-level constraints, not merely design targets [3].

6 Conclusion

Lumped LC and distributed microstrip networks were successfully designed and verified in ADS. The 550 MHz L-network achieved $|S_{11}| = -48.3\ \text{dB}$ (simulated) and $-40.4\ \text{dB}$ (measured), with the 7.9 dB discrepancy attributed to connector parasitics unmodelled in ADS. The 5 GHz network achieved $|S_{11}| = -49.1\ \text{dB}$ with a narrower 6.8% bandwidth due to higher transformation ratio ($Q = 1.34$ vs $Q = 1.0$ at 550 MHz). Lumped matching is practical below $\sim 5\ \text{GHz}$; above this, distributed implementations are preferred to avoid component self-resonance and parasitic degradation.

The Wilkinson divider achieved $|S_{23}| = -24.1\ \text{dB}$ isolation (19 dB improvement with the resistor) and 12.4% return loss bandwidth. The diplexer met all specifications including $\text{VSWR} < 3$ after incorporating $\lambda/4$ isolation lines and arm impedance transformers – both of which were not apparent from the initial analytical design and required iterative simulation.

ADS closed-form models provide fast S-parameter predictions but cannot capture connector parasitics, surface roughness, radiation loss, or higher-order junction modes. Industry-standard processes supplement ADS with full-wave EM solvers (HFSS, CST Microwave Studio) for final validation, particularly for multi-port junctions and tightly coupled structures. The 3.8% line length discrepancy observed in the Wilkinson divider illustrates that Hammerstad formulas are first-order approximations requiring EM-solver correction for precision microwave design. For mmWave applications above 30 GHz, all lumped component designs become infeasible and waveguide or on-chip distributed implementations are required.

References

- [1] D. M. Pozar, *Microwave Engineering*, 4th ed. Hoboken, NJ: Wiley, 2011.
- [2] Keysight Technologies, *Advanced Design System (ADS) 2023*, Santa Rosa, CA, USA, 2023 [Software].
- [3] Microwave101, “Microstrip,” *Microwave Encyclopedia* [Online]. Available: <https://www.microwave101.com>. [Accessed: Mar. 2026].
- [4] M. Wagih, “ENG5056: Lab 3 – Filters, Matching Networks and Resonant Circuits,” University of Glasgow, Glasgow, UK, Lab Sheet, 2023.
- [5] Anthropic, “Claude (claude-sonnet-4-6),” AI language model used for content structuring and $\text{\LaTeX}$ formatting assistance, 2026. Available: <https://www.anthropic.com>.